Practice SOLUTIONS

1. Solve for x on the interval $[0,2\pi]$.

a) $6\sin^2(x)-\sin(x)-1=0$ factor

$(3\sin(x)+1)(2\sin(x)-1)=0$

$3\sin(x)+1=0$

$\sin(x)=-\frac{1}{3}$

R.A.A $=\sin^{-1}\left(\frac{1}{3}\right)\doteq 0.3398$

$x_1=\pi+0.3398\doteq 3.48$

$x_2=2\pi-0.3398\doteq 5.94$

$2\sin(x)-1=0$

$\sin(x)=\frac{1}{2}$

R.A.A $=\pi/6$

$x_3=\pi/6$

$x_4=\pi-\frac{\pi}{6}=\frac{5\pi}{6}$

$\boxed{\therefore x\in\left\{3.48, 5.94, \frac{\pi}{6}, \frac{5\pi}{6}\right\}}$

b) $\cot(x)\cos^2(x)=\cot(x)$

$\cot(x)\cos^2(x)-\cot(x)=0$

$\cot(x)[\cos^2(x)-1]=0$

$\frac{\cos(x)}{\sin(x)}[-\sin^2(x)]=0$

Restriction: $\sin(x)\neq 0 \Rightarrow x\neq k\pi, k\in\mathbb{Z}$

$\cot(x)=0$

$x\in\left\{\frac{\pi}{2},\frac{3\pi}{2}\right\}$

$-\sin^2(x)=0$

$\sin(x)=0$

$x\in\{0,\pi,2\pi\}$ inadmissible

$\boxed{\therefore x\in\left\{\frac{\pi}{2},\frac{3\pi}{2}\right\}}$

c) $4\tan(x)-\sec^2(x)=0$ Restriction: $\cos(x)\neq 0$, $x\neq(2k+1)\frac{\pi}{2}, k\in\mathbb{Z}$

trig. identity

$4\tan(x)-[1+\tan^2(x)]=0$

$-\tan^2(x)+4\tan(x)-1=0$

$\tan^2(x)-4\tan(x)+1=0$

Let $A=\tan(x)$: $A^2-4A+1=0$

$A=\frac{4\pm\sqrt{16-4(1)(1)}}{2(1)}=\frac{4\pm\sqrt{12}}{2}=\frac{4\pm 2\sqrt{3}}{2}=2\pm\sqrt{3}$

$A=2+\sqrt{3}$

$\tan(x)=2+\sqrt{3}$

R.A.A $=\tan^{-1}(2+\sqrt{3})$

$=\frac{5\pi}{12}$

$x_1=\frac{5\pi}{12}$

$x_2=\pi+\frac{5\pi}{12}=\frac{17\pi}{12}$

$A=2-\sqrt{3}$

$\tan(x)=2-\sqrt{3}$

R.A.A $=\tan^{-1}(2-\sqrt{3})$

$=\frac{\pi}{12}$

$x_3=\frac{\pi}{12}$

$x_4=\pi+\frac{\pi}{12}=\frac{13\pi}{12}$

$\boxed{\therefore x\in\left\{\frac{\pi}{12},\frac{5\pi}{12},\frac{13\pi}{12},\frac{17\pi}{12}\right\}}$

d) $2\sin^2(x)-\sin(x)-1=0$ factor

$[2\sin(x)+1][\sin(x)-1]=0$

$2\sin(x)+1=0$

$\sin(x)=-\frac{1}{2}$

R.A.A $=\frac{\pi}{6}$

$x_1=\pi+\frac{\pi}{6}=\frac{7\pi}{6}$

$x_2=2\pi-\frac{\pi}{6}=\frac{11\pi}{6}$

$\sin(x)-1=0$

$\sin(x)=0$

$x_3=\pi/2$

$\boxed{\therefore x\in\left\{\frac{\pi}{2},\frac{7\pi}{6},\frac{11\pi}{6}\right\}}$

e) $4\sin^3(x)+2\sin^2(x)-2\sin(x)-1=0$ factor by grouping

$2\sin^2(x)[2\sin(x)+1]-[2\sin(x)+1]=0$

$[2\sin^2(x)-1][2\sin(x)+1]=0$

$2\sin^2(x)-1=0$

$\sin^2(x)=\frac{1}{2}$

$\sin(x)=\pm\frac{1}{\sqrt{2}}$

R.A.A $=\frac{\pi}{4}$

$x_1=\frac{\pi}{4}$

$x_2=\pi-\frac{\pi}{4}=\frac{3\pi}{4}$

$x_3=\pi+\frac{\pi}{4}=\frac{5\pi}{4}$

$x_4=2\pi-\frac{\pi}{4}=\frac{7\pi}{4}$

$2\sin(x)+1=0$

$\sin(x)=-\frac{1}{2}$

R.A.A $=\frac{\pi}{6}$

$x_5=\pi+\frac{\pi}{6}=\frac{7\pi}{6}$

$x_6=2\pi-\frac{\pi}{6}=\frac{11\pi}{6}$

$\boxed{\therefore x\in\left\{\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4},\frac{7\pi}{6},\frac{11\pi}{6}\right\}}$

f) $\sin(x)+\sqrt{2}=-\sin(x)$

$2\sin(x)+\sqrt{2}=0$

$\sin(x)=\frac{-\sqrt{2}}{2}$

R.A.A $=\frac{\pi}{4}$

$x_1=\pi+\frac{\pi}{4}=\frac{5\pi}{4}$

$x_2=2\pi-\frac{\pi}{4}=\frac{7\pi}{4}$

$\boxed{\therefore x\in\left\{\frac{5\pi}{4},\frac{7\pi}{4}\right\}}$

g) $2\cos(3x-1)=0$, $0 \le x \le 2\pi$

$\cos(3x-1)=0$

Let $A=3x-1$: $\cos A=0$, $-1 \le A \le 6\pi-1$

$A=\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \frac{11\pi}{2}$

$3x-1=\frac{\pi}{2}$, $3x=\frac{\pi}{2}+1$, $x_1=\frac{\pi}{6}+\frac{1}{3}$

$3x-1=\frac{3\pi}{2}$, $x_2=\frac{\pi}{2}+\frac{1}{3}$

$3x-1=\frac{5\pi}{2}$, $x_3=\frac{5\pi}{6}+\frac{1}{3}$

$3x-1=\frac{7\pi}{2}$, $x_4=\frac{7\pi}{6}+\frac{1}{3}$

$3x-1=9\pi/2$, $x_5=\frac{3\pi}{2}+\frac{1}{3}$

$3x-1=\frac{11\pi}{2}$, $x_6=\frac{11\pi}{6}+\frac{1}{3}$

$$\boxed{\therefore x \in \left\{\frac{\pi}{6}+\frac{1}{3}, \frac{\pi}{2}+\frac{1}{3}, \frac{5\pi}{6}+\frac{1}{3}, \frac{7\pi}{6}+\frac{1}{3}, \frac{3\pi}{2}+\frac{1}{3}, \frac{11\pi}{6}+\frac{1}{3}\right\}}$$

h) $\sin(2x)\cos(x)-\cos(2x)\sin(x)=0$

compound angle

$\sin(2x-x)=0$

$\sin(x)=0$

$$\boxed{\therefore x \in \{0, \pi, 2\pi\}}$$

i) $\sec^2(x)-2\tan(x)=4$ Restrictions: $\cos(x) \neq 0$, $x \neq (2k-1)\frac{\pi}{2}$, $k \in \mathbb{Z}$

trig. identity

$\tan^2(x)+1-2\tan(x)=4$

factor

$\tan^2(x)-2\tan(x)-3=0$

$[\tan(x)+1][\tan(x)-3]=0$

$\tan(x)+1=0$

$\tan(x)=-1$

R.A.A $=\pi/4$

$x_1=\pi-\frac{\pi}{4}=\frac{3\pi}{4}$

$x_2=2\pi-\frac{\pi}{4}=\frac{7\pi}{4}$

$\tan(x)-3=0$

$\tan(x)=3$

R.A.A $=\tan^{-1}(3) \doteq 1.249$

$x_3 \doteq 1.25$

$x_4=\pi+1.249 \doteq 4.39$

$\therefore x \in \left\{\frac{3\pi}{4}, \frac{7\pi}{4}, 1.25, 4.39\right\}$

j) $3\tan\left(\frac{x}{2}\right)+3=0$

$\tan\left(\frac{x}{2}\right)=-1$

R.A.A $=\frac{\pi}{4}$

$\frac{x}{2}=\pi-\frac{\pi}{4}$

$\frac{x}{2}=\frac{3\pi}{4}$

$x_1=\frac{3\pi}{2}$

$\frac{x}{2}=2\pi-\frac{\pi}{4}$

$\frac{x}{2}=\frac{7\pi}{4}$

$x=\frac{7\pi}{2}$

inadmissible, $x=\frac{7\pi}{2} \notin [0,2\pi]$

$$\boxed{\therefore x \in \left\{\frac{3\pi}{2}\right\}}$$

k) $-6\sin(2x)\cos(x)+8\cos(2x)+3\sin(x)+4=0$

double angle

$-6[2\sin(x)\cos(x)]\cos(x)+8[1-2\sin^2(x)]+3\sin(x)+4=0$

$\sin^2x+\cos^2x=1$

$-12\sin(x)\cos^2(x)+8-16\sin^2(x)+3\sin(x)+4=0$

$-12\sin(x)[1-\sin^2(x)]-16\sin^2(x)+3\sin(x)+12=0$

$-12\sin(x)+12\sin^3(x)-16\sin^2(x)+3\sin(x)+12=0$

$12\sin^3(x)-16\sin^2(x)-9\sin(x)+12=0$

Factor by grouping

$4\sin^2(x)[3\sin(x)-4]-3[3\sin(x)-4]=0$

$[3\sin(x)-4][4\sin^2(x)-3]=0$

$3\sin(x)-4=0$

$\sin(x)=\frac{4}{3}$

↳ no possible solutions

$4\sin^2(x)-3=0$

$\sin(x)=\pm\frac{\sqrt{3}}{2}$

R.A.A $=\pi/3$

$x_1=\frac{\pi}{3}$

$x_2=\pi-\frac{\pi}{3}=2\pi/3$

$x_3=\pi+\frac{\pi}{3}=\frac{4\pi}{3}$

$x_4=2\pi-\frac{\pi}{3}=\frac{5\pi}{3}$

$$\boxed{\therefore x \in \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\right\}}$$

l) $\cot(x)\cos^2(x)=2\cot(x)$ Restrictions: $\sin(x) \neq 0$, $x \neq k\pi$, $k \in \mathbb{Z}$

$\cot(x)\cos^2(x)-2\cot(x)=0$

$\cot(x)[\cos^2(x)-2]=0$

$\cot(x)=0$

$\frac{\cos(x)}{\sin(x)}=0$

$\cos(x)=0$

$x_1=\frac{\pi}{2}$

$x_2=\frac{3\pi}{2}$

$\cos^2(x)-2=0$

$\underbrace{\cos(x)=\pm\sqrt{2}}_{\text{no possible solutions}}$

$$\boxed{\therefore x \in \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}}$$

m) $\frac{1+\sin(x)}{\cos(x)}+\frac{\cos(x)}{1+\sin(x)}=4$

Restrictions: $\cos(x)\neq 0$ $\quad 1+\sin(x)\neq 0$

$x\neq(2k-1)\frac{\pi}{2},\ k\in\mathbb{Z}$ $\quad \sin(x)\neq -1$

$$\frac{[1+\sin(x)]^2+\cos^2(x)}{[\cos(x)][1+\sin(x)]}=4$$

$$\frac{\sin^2(x)+2\sin(x)+1+\cos^2(x)}{[\cos(x)][1+\sin(x)]}=4$$

$$\frac{2\sin(x)+1+1}{[\cos(x)][1+\sin(x)]}=4$$

$$\frac{2\cancel{[\sin(x)+1]}}{[\cos(x)]\cancel{[1+\sin(x)]}}=4$$

$$\frac{2}{\cos(x)}=4$$

$$\frac{1}{2}=\cos(x)$$

R.A.A $=\pi/3$

$x_1=\frac{\pi}{3}$ $\quad x_2=2\pi-\frac{\pi}{3}=\frac{5\pi}{3}$

$\boxed{\therefore x\in\left\{\frac{\pi}{3},\frac{5\pi}{3}\right\}}$

n) $2\sin^2(x)+3\cos(x)-3=0$

$\sin^2x+\cos^2x=1$

$$2[1-\cos^2(x)]+3\cos(x)-3=0$$

$$2-2\cos^2(x)+3\cos(x)-3=0$$

$$-2\cos^2(x)+3\cos(x)-1=0$$

$$2\cos^2(x)-3\cos(x)+1=0$$

$$[2\cos(x)-1][\cos(x)-1]=0$$

$2\cos(x)-1=0$

$\cos(x)=\frac{1}{2}$

R.A.A $=\pi/3$

$x_1=\frac{\pi}{3}$

$x_2=2\pi-\frac{\pi}{3}=\frac{5\pi}{3}$

$\cos(x)-1=0$

$\cos(x)=1$

$x_3=0$

$x_4=2\pi$

$\boxed{\therefore x\in\left\{0,\frac{\pi}{3},\frac{5\pi}{3},2\pi\right\}}$

o) $2\sin(x)\tan(x)-\tan(x)-2\sin(x)+1=0$

Factor by grouping

$$\tan(x)[2\sin(x)-1]-[2\sin(x)-1]=0$$

$$[2\sin(x)-1][\tan(x)-1]=0$$

$2\sin(x)-1=0$

$\sin(x)=\frac{1}{2}$

R.A.A $=\pi/6$

$x_1=\pi/6$

$x_2=\pi-\frac{\pi}{6}=\frac{5\pi}{6}$

$\tan(x)-1=0$

$\tan(x)=1$

R.A.A $=\frac{\pi}{4}$

$x_3=\frac{\pi}{4}$

$x_4=\pi+\frac{\pi}{4}=\frac{5\pi}{4}$

$\therefore x\in\left\{\frac{\pi}{6},\frac{5\pi}{6},\frac{\pi}{4},\frac{5\pi}{4}\right\}$

p) $\cos(x)\tan(x)-1+\tan(x)-\cos(x)=0$

Restrictions: $\cos(x)\neq 0$

$x\neq(2k-1)\frac{\pi}{2},\ k\in\mathbb{Z}$

Factor by grouping

$$\cos(x)\tan(x)+\tan(x)-\cos(x)-1=0$$

$$\tan(x)[\cos(x)+1]-[\cos(x)+1]=0$$

$$[\cos(x)+1][\tan(x)-1]=0$$

$\cos(x)+1=0$

$\cos(x)=-1$

$x_1=\pi$

$\tan(x)-1=0$

$\tan(x)=1$

R.A.A $=\frac{\pi}{4}$

$x_2=\pi/4$

$x_3=\pi+\frac{\pi}{4}=\frac{5\pi}{4}$

$\therefore x\in\left\{\frac{\pi}{4},\pi,\frac{5\pi}{4}\right\}$

2. A weight hanging from a spring is set in motion moving up and down. Its distance, d, (in cm) above or below its "rest" position is described by $d(t)=5(\sin(6t)-4\cos(6t))$. At what times during the first 2 seconds is the weight at the rest position (d=0).

$$0=5[\sin(6t)-4\cos(6t)]$$

$$0=\sin(6t)-4\cos(6t)$$

$$4\cos(6t)=\sin(6t)$$

$$4=\frac{\sin(6t)}{\cos(6t)}$$

$$4=\tan(6t)$$

Let $A=6t$: $\quad 4=\tan(A)$

R.A.A $=\tan^{-1}(4)\doteq 1.326$

$A\doteq 1.326$	$A\doteq\pi+1.326$	$A\doteq 2\pi+1.326$	$A\doteq 3\pi+1.326$
$6t\doteq 1.326$	$6t\doteq 4.468$	$6t\doteq 7.609$	$6t\doteq 10.751$
$t\doteq 0.22$	$t=0.74$	$t=1.27$	$t=1.79$

$\therefore$ The weight is at the rest position at 0.22, 0.74, 1.27 and 1.79 seconds